

The Australian plague locust, *Chortoicetes terminifera*, is the most important pest species of locust in Australia due to the large areas infested, the frequency of outbreaks and its ability to produce several generations in a year.

[expand all]

Description of Australian plague locust

Adults

Adults of the Australian plague locust can be readily distinguished from other species by the large dark spot on the tip of the hindwings and distinctive scarlet hindleg shanks. Adult body colour is variable and can be grey, brown or green. Adult males measure 25-30 mm long while females are 30-42 mm long. See also: [How to tell the difference between a male and female locust.](#)



Adult Australian plague locust
Dark spot on locust hindwing

Nymphs

The nymphs have five growth stages or instars.



Fifth instar Australian plague locust nymph

First instar nymphs are about 3mm long, pale brown to dark brown or black, and sometimes have a white stripe along the back of its first body segment just behind the head. At each stage the developing wings become more noticeable and can be used to [determine which instar a locust nymph is in](#).



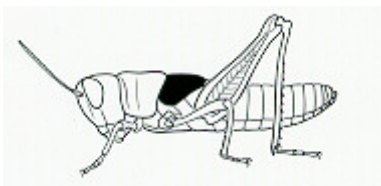
1st instar



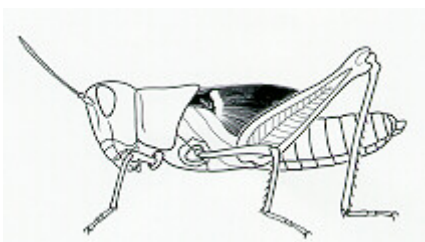
2nd instar



3rd instar 1 cm



4th instar



5th instar

Later instars are grey or brown and sometimes have a white stripe along the back. Further details about this species: [Locust and grasshopper identification guide](#).

Biology and behaviour of the Australian plague locust

Oviposition and Egg Development

Adult locusts feeding on green vegetation can lay their first egg pod 10-14 days after fledging and any subsequent pods can be laid at intervals of 5-10 days in summer and 10-14 days in autumn. Before laying a female may excavate one or more test drill holes to assess soil condition. Once a suitable location is found, a pod of typically 30-60 eggs is laid and sealed with a frothy plug for protection. In the field females usually lay from one to three egg pods. In very dry conditions oocyte development in females can be delayed for many weeks.



egg laying locust

During summer, egg laying usually occurs in areas with some grass cover and egg pods are laid vertically in the soil at a depth of 6-8 cm. In the autumn eggs are often laid in bare soil (frequently on claypans or hard ground along tracks and fences) and many are laid obliquely to the ground surface at a depth of 3-5 cm. Gregarious oviposition often takes place to form dense egg beds, where adult females crowd a given area in response to chemical attractants released by laying females and possibly contained in the secreted froth. Egg pods in these egg beds can reach densities over 500/m² in places.



egg laying swarm (copyright: Paul Zborowski)

Egg development is influenced by both temperature and moisture conditions. Eggs can either develop directly or development can be arrested by quiescence or diapause. Direct development occurs only in warm, moist conditions. The rate of egg development increases with temperature so that complete development can occur in as little as 15 days with a daily maximum of 35°C, while at 25°C it can take a month. Egg development ceases below about 13.5°C and egg death occurs above 38°C. In saturated soils most eggs can survive at least 14 days, unless soil temperature is $\geq 25^{\circ}\text{C}$.

There are several mechanisms which can delay the hatching of eggs of the Australian plague locust. Eggs can undergo one or more of these dormancies at different Embryonic stages of the Australian plague locust, depending on environmental conditions, leading to several possible development pathways.

Egg Diapause

Diapause is a state where growth is suspended by an indirect effect of environmental change to synchronise development to optimal climatic conditions. Diapause occurs in eggs laid during autumn in response to declining daylength experienced by the maternal generation. Diapause occurs in eggs laid from late February to late April in latitudes south of 30°S, with a rapid rise in proportion entering diapause to almost 100% by mid-March. The proportion declines gradually for eggs laid during April, and by May very few eggs enter diapause. Changing photoperiod is the principal stimulus inducing diapause in offspring eggs, but temperature also mediates the proportion eggs entering diapause. Low ($< 15^{\circ}\text{C}$) or high ($> 26^{\circ}\text{C}$) soil temperatures during pre-diapause embryonic stage inhibits diapause. Diapause intervenes when eggs are ~ 40% developed and ends after 7 - 9 weeks. Embryonic development resumes in moist soils at a development rate dependent on soil temperature, but winter temperatures in southern NSW, SA or Victoria delay hatching until October.

Egg Quiescence

Quiescence is the state where embryonic growth is arrested directly by either cold or dry conditions. In dry conditions eggs enter quiescence at either ~ 35% development if the soil was dry at laying, or at ~ 40% if the soil becomes dry while the eggs are in diapause. Dry-induced quiescence can occur at any time of the year, however in summer when ambient temperatures are very high and the soil may

become very dry ($\leq 3\%$ water by weight) most eggs can only survive for 2 – 3 weeks. In cooler periods, most eggs can survive in dry soil for longer due to decreased water loss at lower temperatures.

Nymphs

It takes about 20-25 days for Australian plague locust nymphs (hoppers) to complete development in mid summer. Plague locusts usually have five instars (growth stages) but may have six in dry or cold conditions. Mortality is usually highest during the first instar. Under very dry conditions during summer, high nymphal mortality will occur at any instar. Nymphs are able to withstand cold conditions after emerging and typical winter night-time minimum temperatures are not likely to cause high mortality. Hatchling nymphs are also tolerant to very hot conditions (up to 45 - 53°C depending on the exposure duration) provided that suitable vegetation is present for feeding to replenish water reserves and to provide shelter.

High density nymphs of the Australian plague locust readily form into aggregations called bands. There is coordinated movement of individuals within a band and usually a distinct front develops which can stretch for several kilometres. Bands are rarely more than a few hundred metres deep and the density can range from 1000-5000/m² at the front to less than 50/m² at the rear. Dense bands can often be [seen from the air](#) and resemble a tide mark on the shore.

Nymphs rarely form bands in the first and second instars but may form dense aggregations when basking in the morning sun. Bands are usually not well developed until the third instar and tend to disperse at the fifth. There is often a range of instars within a band. The rate of band movement varies with the band density, the instar, the weather and vegetation cover. Mid-instar bands in dense vegetation may move 50 metres or less per day but late instar bands in can move at 500 metres per day.

Adults

The final moult to the winged adult is called fledging. Development from egg laying to this stage usually takes 7-8 weeks in summer.

The young adult goes through three stages of development

- growth during which the wing muscles are developed and the exoskeleton hardens
- fat accumulation
- oocyte (egg) development.

Each stage can be suppressed if conditions are dry. The growth stage usually lasts about a week. Copulation can occur well before the female starts to develop oocytes and is often not associated with egg laying.

After fledging the adults grow, accumulate fat and often migrate. Lipids are needed as fuel for long distance flight and egg production. If conditions are dry at fledging locusts may not migrate or develop oocytes unless substantial rain falls. If rain does not occur, numbers decline and few adults remain after several months. In some areas conditions may dry off completely during the late instar stage. The locusts generally still fledge but can remain 'papery' and transparent.

Adults vary their behaviour to maintain their body temperature within the range 35-40°C which is the optimum for development, oviposition, flight and feeding. Adults bask when the ambient temperature is low and climb vegetation or seek shade when it is high.

When densities are low adult locusts move short distances by daytime flight. Newly fledged adults often continue to behave as nymphs and move within bands or make very brief low level flights. In gregarious populations the majority of adults fly spontaneously for periods of up to 20 seconds at 2-5 m height. Swarm flight usually only occurs in light winds (<3 m/sec) and at temperatures of between 20°C and 35°C. Swarms generally fly within 15 m of the ground and often appear to roll across the countryside.

The airspeed of freely flying individuals is around 3 metres per second. However even in a strongly flying swarm, a proportion of the locusts are always on the ground feeding or basking and the rate of displacement is thus usually less than the flying speed. The speed and direction of swarm displacement is further modified by hills and trees especially along creek lines which often act as barriers due to the low level flight. Swarm displacement is usually <20 km/day, but may continue for a week or more. Occasionally swarms fly at considerable heights (>30 m) during the day and have been observed from aircraft at heights up to 1,000 m. Such behaviour normally occurs with highly gregarious, very dense populations.

Swarms may persist for many days, but individual swarms often disperse and reform. As a rule, swarms are displaced downwind. However, the locusts usually fly in streams within the swarm and these streams may head in any direction.



locusts taking off at sunset

Migration

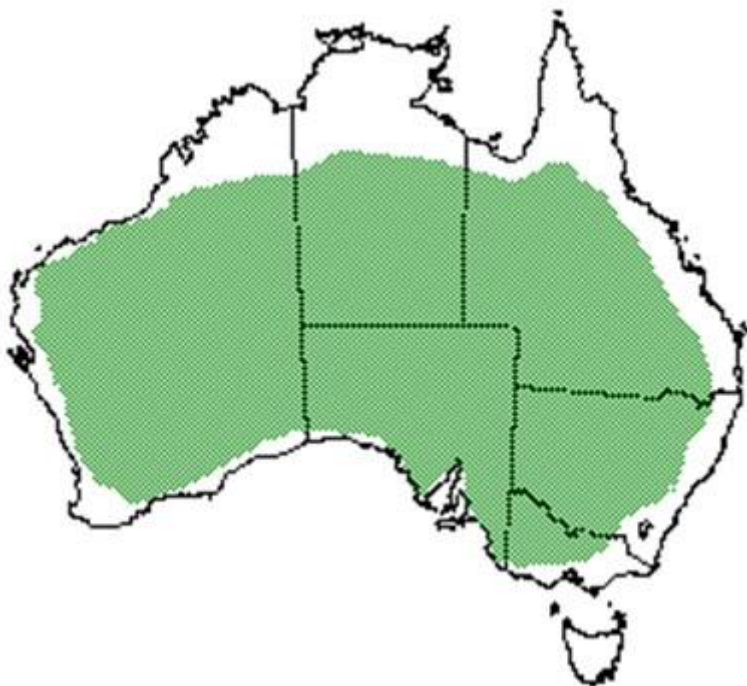
Fledged, pre-reproductive adults often undertake wind-assisted long distance nocturnal migratory flights. Migrations of several hundred kilometres often occur on strong warm winds associated with rain-bearing fronts or low pressure systems. A small proportion of locusts take off individually after sunset on most evenings but when a trough or front is in the area there can be mass take off in groups. The association of mass take off with disturbed weather may increase the chance of locusts reaching rain areas, but does not necessarily result in arrival at destinations suitable for successful breeding (see When locust migration goes wrong).

Night take off is probably stimulated by the decrease in light intensity which is most rapid 20-30 minutes after sunset, which coincides with the period when the plague locusts are usually taking to the air. Take off occurs into the wind and the locusts climb steeply to at least 50 metres height. The locusts

can remain aloft as long as the surface temperature remains above the threshold for flight and land before sunrise, giving a maximum of 9-10 hours displacement. Locusts are often reported as "raining" on rooftops at night.

Observations of night migrations using radar have shown that the locusts usually fly at a height of 300 to 1000 metres. The maximum altitude achieved is probably the height at which the ambient temperature is 20°C (the flight threshold temperature). At such heights the direction and rate of displacement is influenced by the upper level wind flow and the distance travelled depends on the number of hours flown.

Distribution of the Australian plague locust



Map of Australian plague locust distribution

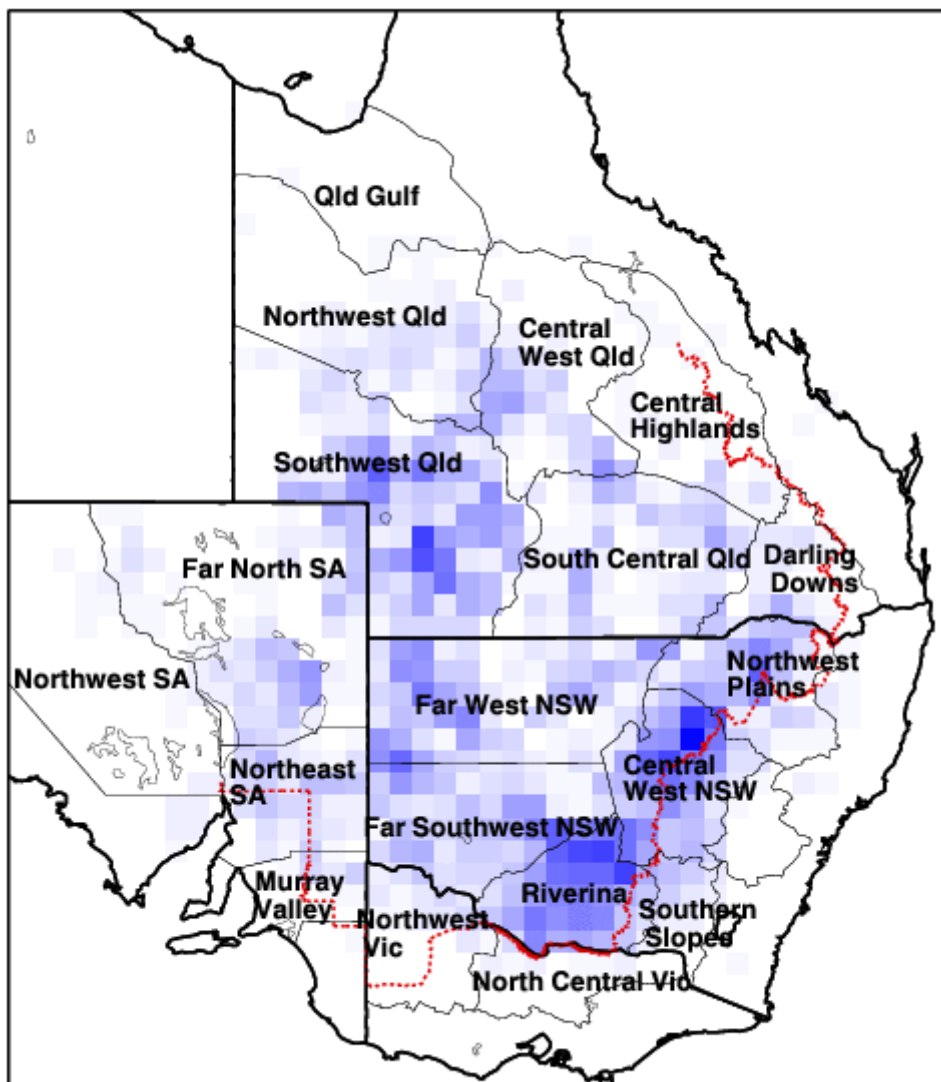
The Australian plague locust is widespread on the mainland and is commonly found in a variety of grassland and open, wooded habitats.

Adult Australian plague locusts are nomadic and swarms can move twenty kilometres in a day. They also make long distance nocturnal migratory flights at heights up to 1000 m. They can be transported over long distances by upper-level winds and are therefore occasionally found in more coastal parts of the mainland and even in northern Tasmania. In eastern Australia migrant locusts sometimes establish populations in valleys east of the Great Dividing Range, such as the Hunter Valley, but these populations usually die out after several generations.

The Australian plague locust is inactive at temperatures below about 15°C and most adults die during the winter months in the southern part of its distribution area. Most eggs laid in autumn in southern areas enter diapause and only resume development in late winter when cold temperatures continue to

delay hatching until spring. Therefore at the start of spring the majority of the population occurs as eggs in the soil which hatch later in spring. The lifecycle of this species gives rise to several generations each year. There are typically three generations corresponding roughly to the spring, summer and autumn seasons, but there may sometimes be two or four depending on latitude and regional conditions.

Within the geographical range of the species in eastern Australia certain regions are subject to frequent locust infestations. Parts of these regions provide favourable habitat for locusts in different seasons. The map below shows the frequency of infestation, estimated by the number of generations during the last 30 years in which high density nymphs or adults have been recorded in 0.5 degree grid cells. During an outbreak high density locusts may occupy an area for several generations. The shading represents the infestation frequency - from white areas with no recorded infestations, through to blue - where high density locusts were recorded in 30 generations. The total number of generations recorded separately during 1977-2008 is 95. Parts of the Riverina, Central West and Far Western NSW, and parts of Southwest Queensland have the highest recorded frequency of infestation. The source of the information is APLC survey and control records.



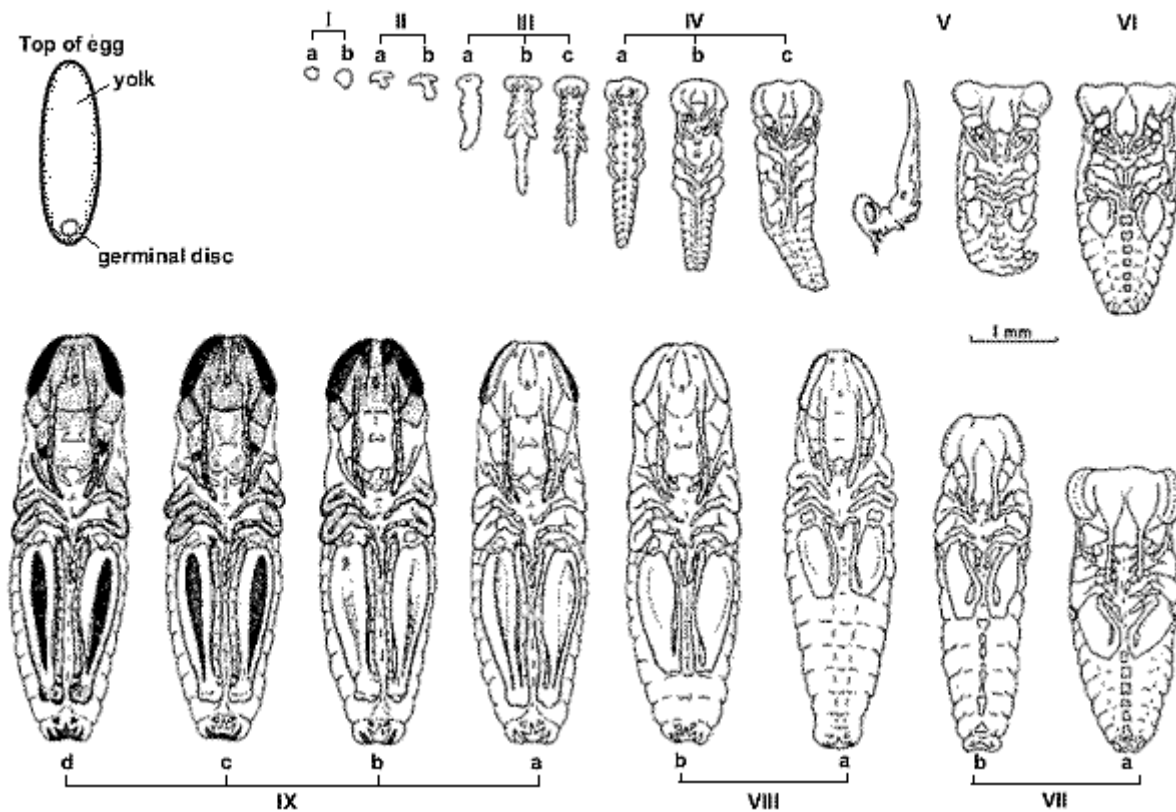
Frequency of locust infestation in eastern Australia from 1977 to 2008

The shading represents the infestation frequency - white areas with no recorded infestations, through to blue - where high density locusts were recorded in 30 generations.

Embryonic stages of the Australian plague locust

Text and diagrams adapted from Wardhaugh (1973) by permission.

Figure 1. Embryonic stages of the Australian plague locust (*Chortoicetes terminifera*). The first illustration shows the entire egg while the other drawings show only the developing embryo.



Stage 0

Period of **blastoderm** development which precedes the appearance of the **germinal disc**.

Stage I (10%)

Formation and growth of the germinal disc at the **micropylar end** of the egg; yolk pale yellow in colour, with fine granular appearance.

- Germinal disc roughly circular in shape.
- Germinal disc pear-shaped but still undifferentiated.

Stage II (15%)

Separation into protocephalic (head) and protocormic (tail) regions but without signs of segmentation.

Embryo contained within the micropylar dome; ratio length/breadth < 2.0.

Protocormic (head) region extending along the ventral side of the egg, the embryo as a whole forming almost a right angle; ratio length/breadth > 2.0.

Stage III (25%)

The first signs of segmentation, and the inward turning of the thoracic appendages are used to define the beginning and end of this stage.

- Gnathal and thoracic appendages present in the form of rounded protuberances.
- Gnathal and thoracic appendages become elongated, segmentation of the anterior margins of the abdomen begins; antennae extend toward anterior margins of mandibular rudiments.
- Gnathal and thoracic appendages project laterally from the embryo; abdominal region greatly extended and representing about half the total length of the embryo; four or five abdominal segments clearly discernible. [Anatreptic movement](#) begins.

Stage IV (30-45%)

This period covers anatrepsis.

- Embryo begins to broaden; appendages become segmented and start to turn inwards and downwards. Development may be arrested at Stage IVa (approximately 30% of complete development) if the soil is dry, i.e. [quiescent](#) stage 1.
- Legs turned in and becoming contiguous along the mid-line, and beginning to retract in the direction of the thorax.
- Hind legs folded into an N-shape; appearance of red-brown pigment in dorsal part of the eye. [Diapause](#) may intervene at Stage IVc (approximately 40-45% of complete development) depending upon environmental conditions during egg maturation. Red brown pigment appears in the dorsal part of the eye in some diapausing eggs. In non-diapause eggs, kept moist, this stage is observed only rarely and the pigmentation of the eye is absent.
The second moisture-induced quiescent stage may intervene at this point if the soil is dry, i.e. [quiescent](#) stage 2. Eggs can survive either in Q1 or Q2 for an entire winter and for 2-3 months in summer. Only a few eggs seem to survive a full summer without rain, but such survival might be important in allowing locusts to survive drought.

Stage V (45-50%)

[Blastokinesis](#) - a rapid process, the initiation being marked by the rupture of the [serosa](#). Embryo arches backwards and rotates around the micropylar end of the egg

Stage VI (55%)

Blastokinesis is complete; the embryo occupies about half the length of the egg, and almost the whole of its width; the dorsal section of the eye becomes strongly pigmented; the remnants of the yolk-plug fill

the anterior half of the egg and extend down its ventral side to the posterior tip of the embryo's abdomen.

Stage VII (60%)

A period of rapid growth during which the yolk is completely engulfed and the embryo extends to fill the entire egg.

- Embryo occupies from half to three-quarters of the egg; the segmentation of antennae and labial palps begins.
- Embryo occupies from three-quarters to the whole of the egg; the dorsal closure, which begins at the tip of the abdomen, is now in progress.

Stage VIII (70%)

The embryo fills the entire egg, dorsal closure is complete, but the engulfed yolk is still clearly visible.

- Hind femur extends to about the fourth abdominal segment and is now shorter than the tibia.
- Hind femur extends to about the eighth abdominal segment; dorsal spurs of the hind tibia are well developed but not yet pigmented.

Stage IX (85%)

The first-instar cuticle is laid down and undergoes a progressive increase in pigmentation prior to hatching.

- Pale orange-brown pigment appears around the median ocellus, along the inner posterior margins of the antennae and on the externo-median area of the hind femur.
- Pigment spreads over the frons, the antennae and the genae, and also develops on the prothoracic and mesothoracic legs; the genitalia and the femoro-tibial articulations also begin to darken. The tibial spurs are developed but not yet pigmented.
- Manibular cusps and tibial spurs begin to darken.
- Tibial spines and tarsal spurs darken; the embryo is ready to hatch.

Glossary

Anatrepsis/anatreptic movement: movement of the embryo away from the posterior pole of the egg.

Blastoderm: layer of cells that surround the yolk at the early stage of embryonic development.

Blastokinensis: active movement of embryo where it passes from the ventral to the dorsal side of the egg and at the same time revolves 180 degrees on its long axis.

Diapause: a state where growth is arrested by an indirect effect of the environment. In the Australian plague locust, eggs laid during the decreasing daylengths of autumn may enter diapause at late anatrepsis when temperatures are mild (less than 25 degrees Celsius) but not hot (greater than 32

degrees Celsius). Diapause results in adult locusts not being present during the winter when it is too cold in temperate areas and too dry in subtropical areas for growth. Diapause ends 9 (+ 2) weeks later (usually by mid-June) and development resumes as soon as moisture and temperature conditions are suitable.

Germinal disc: as a result of increased cell division, the blastoderm (see above) becomes thicker in the ventral region of the egg. This thickening is called the germ band, which develops into the future embryo. In the locust egg the germ band initially takes on the form of a small disc before becoming differentiated into a primary 'head' and 'tail' region.

Micropylar end: refers to the end of the egg where the micropyle, a specialised pore, is found.

Quiescence: a state where growth is arrested directly by either cold or dry conditions. Unlike diapause, growth resumes as soon as conditions are favourable. In the Australian plague locust, embryos absorb most of their moisture between 25 and 45% development and if the soil is dry at the time of laying, no moisture is available for development to proceed and growth is arrested near the beginning of the moisture absorption stage i.e. at about 30% development. This arrest at 30% development is called quiescent stage 1 or Q1. Embryos can also enter quiescence after diapause is complete. During diapause, development is arrested at late anatrepsis (40-45% development) and if it is dry at the end of diapause, development is arrested at quiescent stage 2 or Q2 and does not resume until rain falls. Eggs can survive in a state of quiescence over an entire winter and for 2-3 months during summer. Only a few seem able to survive a full summer without rain, but such survival may be important in allowing locusts to survive drought.

Serosa: layer of cuticle formed around the yolk by the blastoderm during the early stages of embryonic development.

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Further reading

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Photos of locust egg laying, egg pods and hatchings

Photographs of Australian plague locust (*Chortoicetes terminifera*) egg laying, egg pods and emerging nymphs.



Egg laying swarm of Australian plague locusts

Photo courtesy R. Eade, NSW LHPA



Exposed locust egg pod with white froth plug at soil surface
The curved shape is typical of shallow autumn laid egg pods



Locust nymphs (3-4 mm long) emerging from soil.
The white 'caps' are the remains of the egg yolk sac.

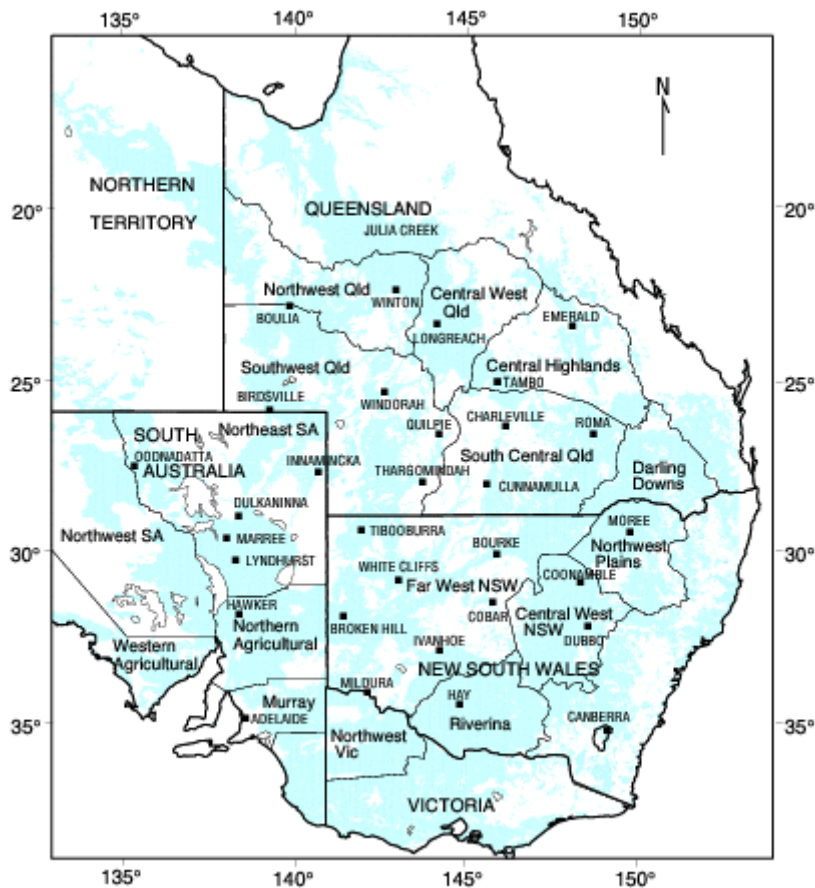
Also see:

Locust habitats

The potential habitats of the Australian plague locust cover half of inland eastern Australia, an area of about two million square kilometres. They are primarily open tussock grasslands on clay, loam or stone-mantled desert loam soils. Locust habitats include the Mitchell grass downs of western Queensland and the chenopod low open shrublands across southern Australia. Habitats become suitable for locust breeding after rainfall, when soil moisture allows egg development and vegetation response provides food for subsequent survival of nymphs.

Landscapes which are generally unfavourable for locust breeding include forest, woodland, rocky hills, desert sandplain and dunefields. The clearing of forest and woodland vegetation on clay and loam soils has expanded the area of potential habitat in the agricultural regions of southern and eastern Australia. A map based on soil types and subdivided in areas where tree clearing has produced some areas of grassy vegetation, including pasture and agricultural landscapes, accounts for more than 80 per cent of all historical observations of high density locusts.

The map below shows the geographical extent of potential habitat for Australian plague locust. These landscapes are also the habitats of spur-throated and migratory locusts, particularly in the tropics.



Within the regions of potential habitat there are some areas which are subject to more frequent infestation. Female Australian plague locusts typically lay their eggs in hard-packed soil along roads and tracks, in clay pans or in stony areas, such as those shown below.



Stony Mitchell Downs, near Eromanga, Queensland